

EF Core in ASP.NET



1. Exercise Overview

In this lab, you will work on a small ASP.NET Core MVC application called **MoviesApp**. The project is already created and partially implemented.

Your main tasks are:

- to **write EF Core queries** in the service layer
- to **connect controllers with services**
- to **implement data import** from JSON and XML files

The focus of the exercise is **working with the database through EF Core**, not on HTML/CSS or advanced ASP.NET features.

2. Project Structure

The solution contains a single ASP.NET Core MVC project:

- **Controllers**
 - MoviesController
 - WatchlistController
 - ImportController
- **Data**
 - ApplicationDbContext – EF Core DbContext
 - DataSets/moviesJSONFile.json – sample data for JSON import
 - DataSets/MoviesXMLFile.xml – sample data for XML import
- **Models**
 - Movie – entity for a movie
 - Watchlist – entity that connects users with movies (many-to-many)
- **Services**
 - Interfaces/IImportService
 - Interfaces/IMoviesService
 - Interfaces/IWatchlistService
 - ImportService

- MoviesService
- WatchlistService
- **ViewModels**
 - ViewModels/Movies – view models for listing, adding, and showing details for movies
 - ViewModels/Import – view model for the Import index page
- **Views**
 - Views/Movies – pages for All Movies, Add Movie, Details, etc.
 - Views/Watchlist – pages for the watchlist
 - Views/Import/Index.cshtml – page with buttons for importing from JSON / XML

3. Learning Objectives

After finishing the exercise, you should be able to:

1. Use **DbContext** in a service class to query and modify data.
2. Write **LINQ queries** with Where, Select, OrderBy, Include, etc.
3. Map entity data to **ViewModels** (and later to DTOs).
4. Use **dependency injection** to inject services into controllers.
5. Implement **import** logic from JSON and XML files into the database.

4. Explore and Run the Application

Open the solution **MoviesApp** in Visual Studio.

Inspect the **project structure** in Solution Explorer and find the folders listed above.

Open **AppDbContext** and check:

- registered DbSets
- OnModelCreating configuration if available.

Make sure the database exists:

- Run Update-Database from **Package Manager Console**, or
- Use dotnet ef database update from the terminal.

You will notice:

- The application is **fully functional from a UI perspective**
- The pages **load correctly** and navigation works
- Some pages (Movies, Watchlist, Import) show **dummy data**, which is hard-coded

At this stage, **no real database queries are executed** because:

- **The services are still empty**
- **The controllers do not call the services**
- Movies and watchlist content come from temporary in-memory lists or placeholders

Your task in this exercise will be to **replace all dummy data with actual EF Core database queries**.

5. Implementation Steps

In the next part of the exercise, you will gradually **replace the dummy data with real database logic by completing the services and connecting them to the controllers**.

Follow the steps below in order:

- Implement the **MovieService** (EF Core Queries)
- Connect **MoviesController** with **MovieService**
- Implement the **WatchlistService** (EF Core Queries)
- Connect **WatchlistController** with **WatchlistService**
- Prepare DTOs for Import (**JSON & XML**)
- Implement **ImportService**
- Connect **ImportController** with **ImportService**

6. Testing Your Work

1. Start the application and open the **Import** page.
2. Try importing from **JSON** and **XML**:
 - Check the message for the count of imported movies.
 - Verify the movies are really stored in the database (All Movies page and database).
3. Open the **Movies** section:
 - Check that all movies are listed correctly.
 - Test the **Add / Details / Edit / Delete** operations (depending on what is implemented).
4. Open the **Watchlist** section:
 - Add movies to your watchlist and check if they appear correctly.
 - Test removing movies from the watchlist.

7. Final Version

At the end of the exercise, you should have:

- Fully implemented **MoviesService**, **WatchlistService**, and **ImportService**.
- **Controllers using their services**.

Working pages for:

- All Movies
- Add Movie
- Movie Details
- Watchlist
- Import (JSON and XML)

All EF Core queries written by you.